

Ashima Arora

ashimaarora63@gmail.com | (929) 369-9623 | Colonia, NJ | [LinkedIn](#) | [Portfolio](#) | [Github](#)

SUMMARY

Data Engineer with **4+ years of experience** in designing and optimizing scalable data pipelines, now transitioning into **Machine Learning Engineer & MLOps**. Proficient in **Python, distributed computing frameworks, and cloud services**, with hands-on experience in **research and development of ML models for predictive analytics, Natural Language Processing, Deep Learning, and recommendation systems**. Skilled in **model evaluation, feature engineering, and operationalizing ML pipelines**. Experienced in **Agile, SDLC, and technical documentation** to ensure collaboration and high-quality deliverables.

TECHNICAL SKILLS

Programming & Data: Python, Java, SQL, NoSQL databases, Hadoop, Hive, Spark/PySpark, Kafka, Tableau, Shell Scripting
Machine Learning Libraries/Frameworks: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn
Machine Learning Models: Regression, Classification, Clustering, Dimensionality Reduction, Principal Component Analysis (PCA), Decision Trees, Random Forest, Gradient Boosting, CNN, RNN, LSTM.
Cloud Computing & Tools: Google Cloud Platform, Kubernetes, Cloud Storage, BigQuery, DataProc, Compute Engine, AWS Glue, AWS Athena, AWS ML Services, Airflow, Jenkins, UrbanCode Deploy, Docker, Continuous Integration / Continuous Delivery (CI/CD), Git, GitHub, Monitoring Tools
Collaboration & Process: Agile methodologies, Software development lifecycle, Technical Documentation, Process Standardization, Knowledge Sharing, Mentoring, Cross Functional Communication, Operational Excellence
Others: Linux, Confluence, JIRA, API Integration and Development, Data Modeling, Data Migration, Data Quality Assurance

EDUCATION

- The University of Texas at Austin - Master of Science in Computer Science, Aug 2022 - 2024; Focus in Machine Learning & AI*
- CUNY Queens College - Bachelor of Science in Computer Science, Aug 2017 - June 2021*

WORK EXPERIENCE

Software Engineer - Big Data | Wells Fargo | June 2021 to Present

- Engineered 8 scalable data pipelines using Hadoop, Hive, and Spark (PySpark), supporting **batch and near real-time data processing** of up to **10 TB of data monthly** and improving **data pipeline efficiency by 25%**.
- Execute **data integration strategies** to unify over 10 disparate internal and external data sources, enabling a **60% improvement in data accessibility** and analytical consistency.
- Co-architect **cloud based data solutions** to improve data accessibility and enable scalable analytics solutions.
- Streamlined and **automated complex ETL workflows** using **Python and Autosys**, **increasing operational efficiency by 40%** by reducing manual effort and improving data reliability and accuracy.
- Integrated Apache Kafka** to enable high-throughput, **real-time data ingestion**, **increasing data processing speed by 35%** and **maintaining integration documentation** to support **production stability and knowledge transfer** across the team.
- Designed and prototyped **AI agent-based internal automation workflows** using **Microsoft Copilot Studio**, integrating LLM-powered workflows with enterprise knowledge bases and structured data sources.
- Lead a team of 5 engineers** through **Agile sprint cycles**, facilitating **cross-functional collaboration** with product managers, analysts, and DevOps to deliver reliable data pipeline and visualization solutions on time.
- Spearheaded performance optimization initiatives, **reduced product average feature time to market by 40%**, while continuing to keep the total production defect rates below organization benchmarks.

Research Intern | Research Foundation of CUNY | August 2019 to November 2019

- Designed a feedback pipeline** to capture and categorize user input (bug reports, feature requests) from testers and panelists, storing data in **Excel/ SQL for structured analysis** and **maintaining clear documentation of findings** for engineering teams.
- Performed **exploratory data analysis (EDA)** on feedback trends to identify high-impact issues and **standardizing prioritization criteria** for feature development, **reducing time-to-fix for critical bugs by 30%**.
- Streamlined participant recruitment tracking and built dashboards** to monitor response rates, increasing participation by **35%**.

KEY PROJECTS

Object Detector and Auto-Pilot Driving — *Python, PyTorch, Fully Convolutional Neural Network (CNN)*

- Developed an **object detection program** for **auto-pilot driving in a simulated environment**, using **PyTorch** and **Fully Convolution Neural Networks (FCN)**, achieving over **90% detection accuracy** and enabling **reliable real-time decision-making**.

Electricity Consumption & Pricing Prediction — *Python, scikit-learn, Pandas, NumPy*

- Conducted **correlation analysis** to identify key drivers of electricity demand and pricing; identifying **generation metrics** as stronger predictors than weather-related features.
- Built and evaluated **forecasting models (Random Forest, Gradient Boosting, RNN, LSTM)** against a **Linear Regression baseline**, achieving lowest **RMSE** for demand with Random Forest and for price with LSTM (69% lower vs. baseline).

Movie Recommendation System — *Python, scikit-learn, Cosine Similarity, Fuzzy Matching*

- Built a hybrid movie recommender system that uses **popularity-based filtering** and **content-based filtering** using **cosine similarity**.
- Filtered movies on **genres, tagline, keywords, director, cast, and title** to generate **highly relevant recommendations**, and implemented **fuzzy matching** to correct typos in movie titles and improve user experience.

Sentiment Analysis for Amazon's Product Reviews — *Python, Natural Language Toolkit (NLTK)*

- Built a **Natural Language Processing (NLP)** pipeline to preprocess and **classify Amazon Alexa reviews by sentiment** using **Random Forest and XGBoost**, achieving **94% test accuracy** and generating **data driven insights from customer feedback**.
- Applied feature engineering, preprocessing, and visualization to extract actionable insights for product improvement.

CERTIFICATIONS

Google Cloud Certified – Associate Cloud Engineer, AWS Machine Learning - Specialty (In Progress)